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HW01

실습 1: 기본 내비게이션 (AMCL + 기본 지도)

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목차

1. 들어가며......2

2. 실행 결과 및 고찰......2

2.1 실행 결과 자료......2

2.2 약간의 고찰......3

2.2.1 Managed nodes are active......3

2.2.2 2D Pose Estimate......4

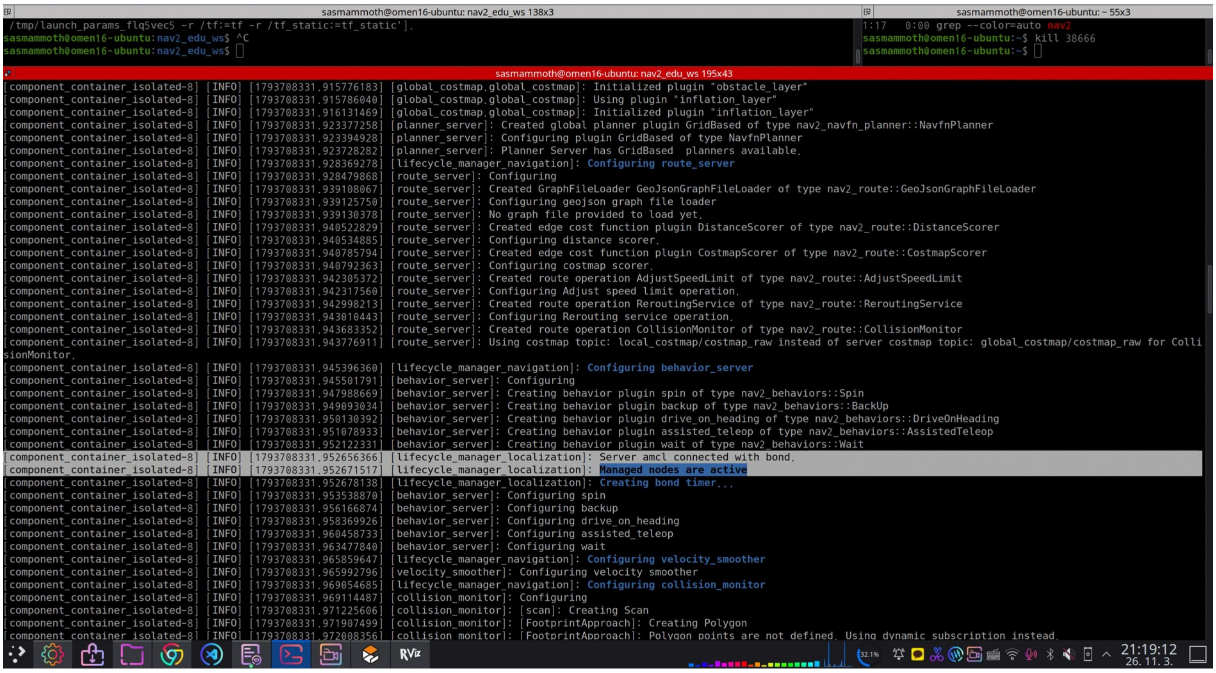
1. 들어가며

이 과제는 RViz2 와 Gazebo 를 처음 실행하여 보고 2D Pose Estimate 등을 이용해 로봇의 시작점을 세팅한다. 터미널에 Managed nodes are active 를 확인한다.

2. 실행 결과 및 고찰

2.1 실행 결과 자료

영상 링크 : https://drive.google.com/file/d/1bTPU02KuYEXFHNh4w9HTSzTwMZADtedN/view?usp=sharing

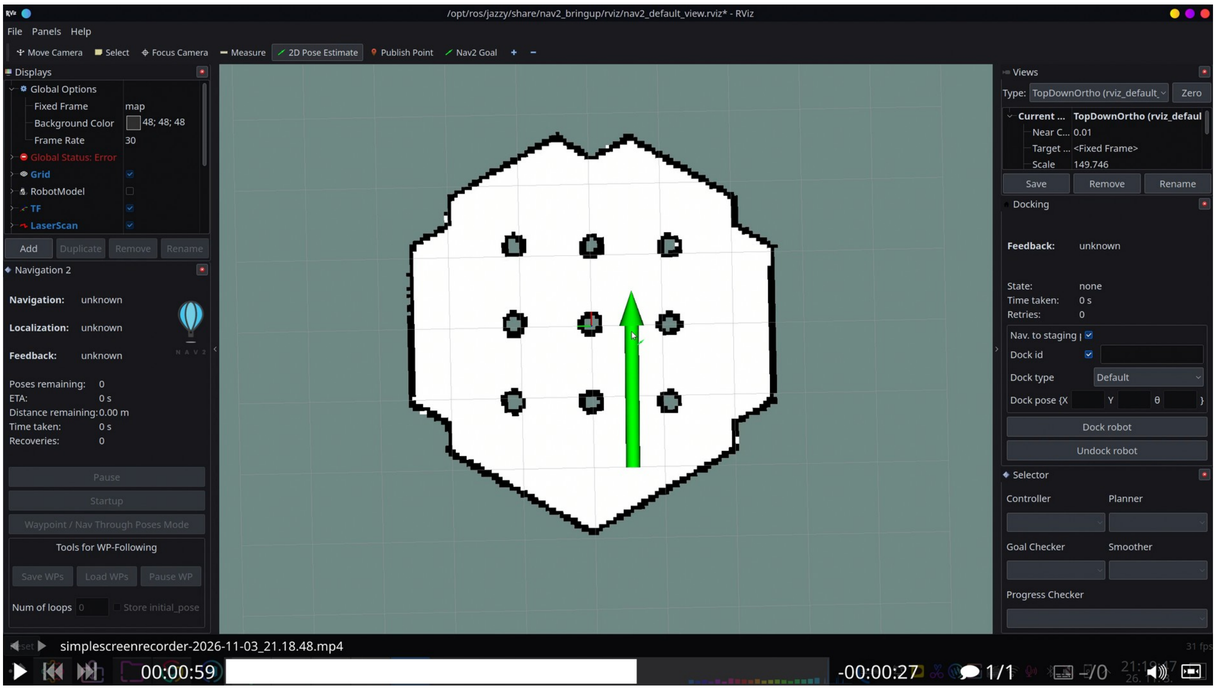

sasmammoth@omen16-ubuntu: nav2_edu_ws 138x3
sasmammoth@omen16-ubuntu: ~ 55x3
sasmammoth@omen16-ubuntu: nav2_edu_ws 195x43
RViz
21:19:12
26. 11. 3.

```
/tmp/launch_params_flq5vec5 -r /tf:=tf -r /tf_static:=tf_static'].
sasmammoth@omen16-ubuntu:nav2_edu_ws$ ^C
sasmammoth@omen16-ubuntu:nav2_edu_ws$ 
```

```
1:17   0:00 grep --color=auto nav2
sasmammoth@omen16-ubuntu:~$ kill 38666
sasmammoth@omen16-ubuntu:~$ 
```

```
[component_container_isolated-8] [INFO] [1793708331.915776183] [global_costmap.global_costmap]: Initialized plugin "obstacle_layer"
[component_container_isolated-8] [INFO] [1793708331.915786040] [global_costmap.global_costmap]: Using plugin "inflation_layer"
[component_container_isolated-8] [INFO] [1793708331.916131469] [global_costmap.global_costmap]: Initialized plugin "inflation_layer"
[component_container_isolated-8] [INFO] [1793708331.923377258] [planner_server]: Created global planner plugin GridBased of type nav2_navfn_planner::NavfnPlanner
[component_container_isolated-8] [INFO] [1793708331.923394928] [planner_server]: Configuring plugin GridBased of type NavfnPlanner
[component_container_isolated-8] [INFO] [1793708331.923728282] [planner_server]: Planner Server has GridBased  planners available.
[component_container_isolated-8] [INFO] [1793708331.928369278] [lifecycle_manager_navigation]: Configuring route_server
[component_container_isolated-8] [INFO] [1793708331.928479868] [route_server]: Configuring
[component_container_isolated-8] [INFO] [1793708331.939108067] [route_server]: Created GraphFileLoader GeoJsonGraphFileLoader of type nav2_route::GeoJsonGraphFileLoader
[component_container_isolated-8] [INFO] [1793708331.939125750] [route_server]: Configuring geojson graph file loader
[component_container_isolated-8] [INFO] [1793708331.939130378] [route_server]: No graph file provided to load yet.
[component_container_isolated-8] [INFO] [1793708331.940522829] [route_server]: Created edge cost function plugin DistanceScorer of type nav2_route::DistanceScorer
[component_container_isolated-8] [INFO] [1793708331.940534885] [route_server]: Configuring distance scorer.
[component_container_isolated-8] [INFO] [1793708331.940785794] [route_server]: Created edge cost function plugin CostmapScorer of type nav2_route::CostmapScorer
[component_container_isolated-8] [INFO] [1793708331.940792363] [route_server]: Configuring costmap scorer.
[component_container_isolated-8] [INFO] [1793708331.942305372] [route_server]: Created route operation AdjustSpeedLimit of type nav2_route::AdjustSpeedLimit
[component_container_isolated-8] [INFO] [1793708331.942317560] [route_server]: Configuring Adjust speed limit operation.
[component_container_isolated-8] [INFO] [1793708331.942998213] [route_server]: Created route operation ReroutingService of type nav2_route::ReroutingService
[component_container_isolated-8] [INFO] [1793708331.943010443] [route_server]: Configuring Rerouting service operation.
[component_container_isolated-8] [INFO] [1793708331.943683352] [route_server]: Created route operation CollisionMonitor of type nav2_route::CollisionMonitor
[component_container_isolated-8] [INFO] [1793708331.943776911] [route_server]: Using costmap topic: local_costmap/costmap_raw instead of server costmap topic: global_costmap/costmap_raw for Colli
sionMonitor.
[component_container_isolated-8] [INFO] [1793708331.945396360] [lifecycle_manager_navigation]: Configuring behavior_server
[component_container_isolated-8] [INFO] [1793708331.945501791] [behavior_server]: Configuring
[component_container_isolated-8] [INFO] [1793708331.947988669] [behavior_server]: Creating behavior plugin spin of type nav2_behaviors::Spin
[component_container_isolated-8] [INFO] [1793708331.949093034] [behavior_server]: Creating behavior plugin backup of type nav2_behaviors::BackUp
[component_container_isolated-8] [INFO] [1793708331.950130392] [behavior_server]: Creating behavior plugin drive_on_heading of type nav2_behaviors::DriveOnHeading
[component_container_isolated-8] [INFO] [1793708331.951078933] [behavior_server]: Creating behavior plugin assisted_teleop of type nav2_behaviors::AssistedTeleop
[component_container_isolated-8] [INFO] [1793708331.952122331] [behavior_server]: Creating behavior plugin wait of type nav2_behaviors::Wait
[component_container_isolated-8] [INFO] [1793708331.952656366] [lifecycle_manager_localization]: Server amcl connected with bond.
[component_container_isolated-8] [INFO] [1793708331.952671517] [lifecycle_manager_localization]: Managed nodes are active
[component_container_isolated-8] [INFO] [1793708331.952678138] [lifecycle_manager_localization]: Creating bond timer...
[component_container_isolated-8] [INFO] [1793708331.953538870] [behavior_server]: Configuring spin
[component_container_isolated-8] [INFO] [1793708331.956166874] [behavior_server]: Configuring backup
[component_container_isolated-8] [INFO] [1793708331.958369926] [behavior_server]: Configuring drive_on_heading
[component_container_isolated-8] [INFO] [1793708331.960458733] [behavior_server]: Configuring assisted_teleop
[component_container_isolated-8] [INFO] [1793708331.963477840] [behavior_server]: Configuring wait
[component_container_isolated-8] [INFO] [1793708331.965859647] [lifecycle_manager_navigation]: Configuring velocity_smoother
[component_container_isolated-8] [INFO] [1793708331.965992796] [velocity_smoother]: Configuring velocity smoother
[component_container_isolated-8] [INFO] [1793708331.969054685] [lifecycle_manager_navigation]: Configuring collision_monitor
[component_container_isolated-8] [INFO] [1793708331.969114487] [collision_monitor]: Configuring
[component_container_isolated-8] [INFO] [1793708331.971225606] [collision_monitor]: [scan]: Creating Scan
[component_container_isolated-8] [INFO] [1793708331.971907499] [collision_monitor]: [FootprintApproach]: Creating Polygon
[component_container_isolated-8] [INFO] [1793708331.972008356] [collision_monitor]: [FootprintApproach]: Polygon points are not defined. Using dynamic subscription instead.
```

그림 1. (24 초 부근) ‘localization’ 이 active 되었음



/opt/ros/jazzy/share/nav2_bringup/rviz/nav2_default_view.rviz* - RViz
File Panels Help
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
Publish Point
Nav2 Goal
Displays
Add
Duplicate
Remove
Rename
Navigation 2
NAV2
Pause
Startup
Waypoint / Nav Through Poses Mode
Tools for WP-Following
Save WPs
Load WPs
Pause WP
Views
Zero
Save
Remove
Rename
Docking
Dock robot
Undock robot
Selector
Controller
Planner
Goal Checker
Smoother
Progress Checker
simplescreenrecorder-2026-11-03_21.18.48.mp4
00:00:59
-00:00:27
1/1
31 fps

Global Options
Fixed Frame: map
Background Color: 48; 48; 48
Frame Rate: 30
Global Status: Error
Grid ☑
RobotModel ☐
TF ☑
LaserScan ☑

Navigation: unknown
Localization: unknown
Feedback: unknown

Poses remaining: 0
ETA: 0 s
Distance remaining: 0.00 m
Time taken: 0 s
Recoveries: 0

Num of loops 0 ☐ Store initial_pose

Type: TopDownOrtho (rviz_default_
Current ...: TopDownOrtho (rviz_defaul
Near C...: 0.01
Target ...: <Fixed Frame>
Scale: 149.746

Feedback: unknown

State: none
Time taken: 0 s
Retries: 0

Nav. to staging ☑
Dock id ☑
Dock type: Default
Dock pose {X Y θ }

그림 2. (58초 부근) 2D Pose Estimate를 이용하여 초기 위치를 찍음

2.2 약간의 고찰

2.2.1 Managed nodes are active

프로그램을 실행하고 초기 위치를 찍으면 ‘Managed nodes are active’ 라는 로그가 나온다. 이는 수업 때 잠깐 말하고 간 RViz2가 무겁기 때문에 노드마다 라이프사이클을 달아서 순서대로 실행하고 종료되도록 한다고 한 것과 연관이 있는거 같다.

2.2.2 2D Pose Estimate

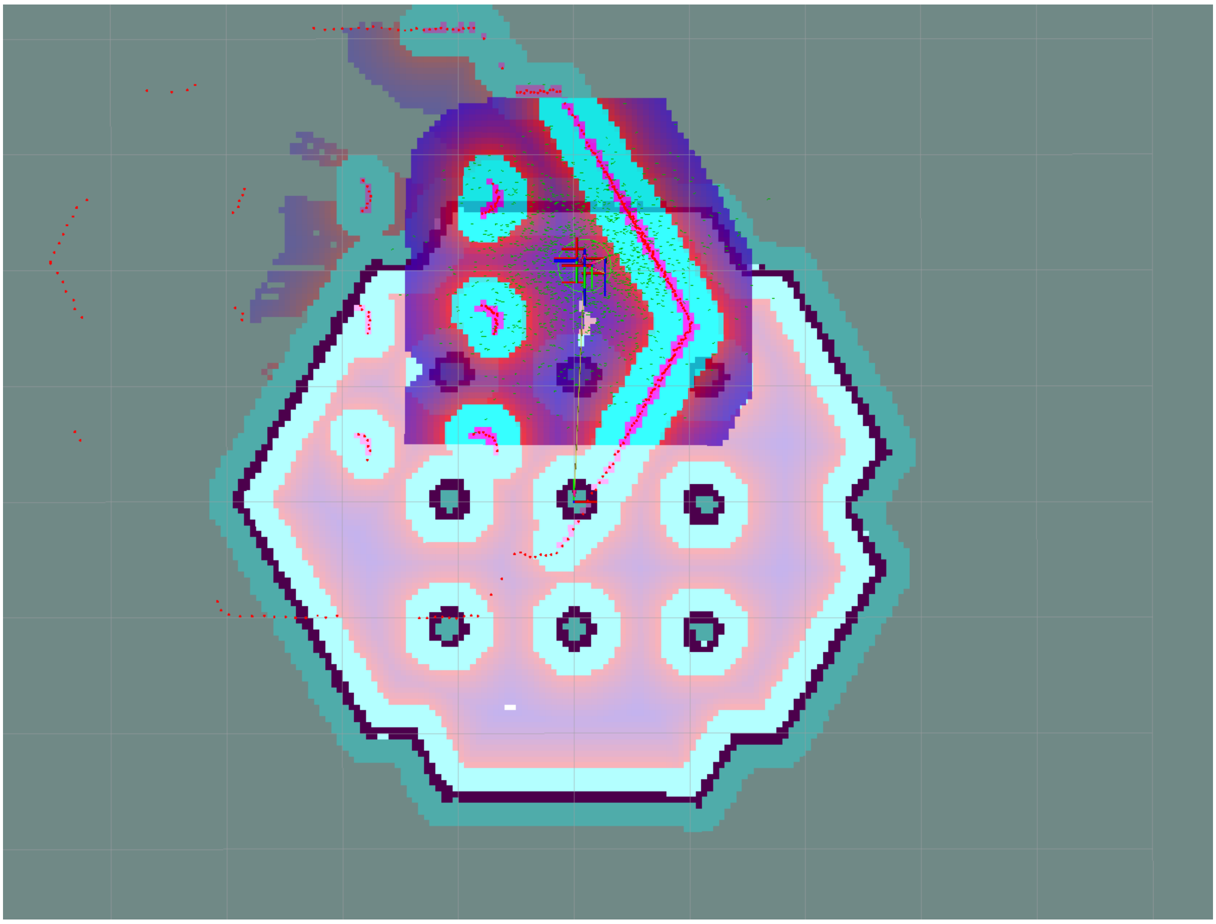


그림 3. 멀리 떨어진 아무 위치를 클릭하였을때

처음 초기 위치를 자동으로 가져오지 못하여 Gazebo 의 위치를 보고 위치를 선정해주어야 한다. 2D pose Estimate 를 하면 현재 위치를 클릭한 위치라고 생각하고 다음 명령을 실행하기 때문에 실제 맵과 라이다로 받아온 벽들과 차이가 있을 수 있다. 이를 보정하는 것은 HW02 에서 한다.